

Post-Wardropian traffic assignment framework for fleets of autonomously routing vehicles

Jagiellonian University, Kraków
Rafał Kucharski
rafal.kucharski@uj.edu.pl
<https://rafalkucharskilab.pl/>



Motivation



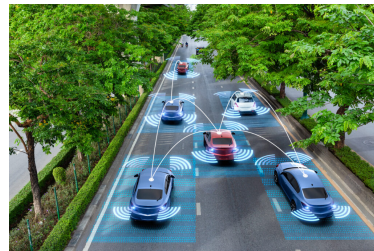
Overview

Connected autonomous vehicles will **soon** be able to apply arbitrary routing strategies:

- not only behavioural decisions of random utility maximisers,
- but also **collective** decisions given an **arbitrary** objective.

Such a route choice process will not fit into the well-established Wardropian paradigm of traffic assignment.

We introduce a Post-Wardropian framework to better model, understand, predict, and control fleet routing behaviour in future CAV cities.



The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.



The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.

The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.



The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.

The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.

The future

- Human drivers make individual routing decisions and, despite variability in supply and demand, their routing behaviour is somehow stable.
- CAV fleet providers offer autonomous routing services which, at first, are not different from human routing. Some drivers join the fleet and the operator increases its market share.

now

future

- Beyond a critical mass, the fleet operator may assign controlled vehicles to arbitrary routes for its own benefit.
- The operator develops a routing strategy optimising some latent goal, e.g. profit or market share.
- New fleet operators may enter and compete with human drivers and other fleets, with tailored pricing and routing strategies.
- The city may eventually regulate undesired fleet actions and their negative impact.

Missing framework

A unified framework to simulate such scenarios and understand their impact is currently missing.

Wardropian baseline



Wardropian user equilibrium

Let's consider a traffic demand Q on a road network:

$$Q_i = \{o_i, d_i, \tau_i, \theta_i\}.$$

Every day each individual driver i , based on personal traits, knowledge and experience θ_i , **selects** a path from her choice set K_i :

$$k_i = \arg \max_{k \in K_i} u_{k,i}$$

- $u_{k,i}$ - expected path utility, typically $u_{k,i} \propto t_{k,i}$ or cost $c_{k,i}$.
- choices (joint action) determine network travel times

$$t = \mathcal{M}(\mathcal{K}).$$

from some traffic model (BPR, MFD, Microsimulation, ...)

Day-to-day process

Drivers **learn** from experienced travel times, update expectations and adapt route choices.

Post-Wardropian



What changes with CAV fleets?

The fleet operator controls:

1 **choice set**

$$K_i \rightarrow K_i^*$$

2 **utility**

$$u_{k,i} \rightarrow u_{k,i}^*$$

3 **travel-time prediction**

$$t_k \rightarrow t_k^*$$

4 **routing / assignment**

$$k_i = \arg \max_{k' \in K_i} u_{k',i} \rightarrow k_i^*$$



1. Choice set

From K_i to K_i^*

Route guidance can already suggest paths which would not be considered by the driver.

However,

- CAV user may accept more exotic paths (Uber driver case).
- CAVs may therefore have larger choice sets.
- This extends the search space and more complex network loading patterns may emerge.

This will only slightly affect the system



2. Arbitrary utility

Wardropian Utility

$$u_{k,i} \propto t_{k,i}$$

Not anymore

Utility is under the control of the fleet operator and may include:

- $t(k_i)$ — own travel time,
- $T(k_i)$ — system-wide total travel time,
- $T_F(k_i)$ — total fleet travel time,
- $T_{-F}(k_i)$ — total travel time outside the fleet.

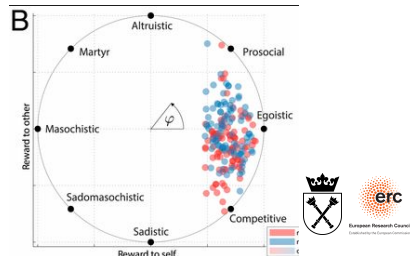
General CAV path utility

$$u_{k,i} = - \left(\gamma_i t(k_i) + \gamma_T T(k_i) + \gamma_F T_F(k_i) + \gamma_{-F} T_{-F}(k_i) \right)$$

Routing behaviour

Routing depends on the γ parameters:

- **Egoistic:** $\gamma_i = 1$
- **Social:** $\gamma_T = 1$
- **Selfish fleet:** $\gamma_F = 1$
- **Antisocial / competitive:** $\gamma_T, \gamma_{-F} < 0$



2. Arbitrary utility

Wardropian Utility

$$u_{k,i} \propto t_{k,i}$$

Not anymore

Utility is under the control of the fleet operator and may include:

- $t(k_i)$ — own travel time,
- $T(k_i)$ — system-wide total travel time,
- $T_F(k_i)$ — total fleet travel time,
- $T_{-F}(k_i)$ — total travel time outside the fleet.

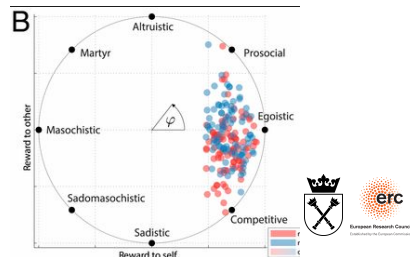
General CAV path utility

$$u_{k,i} = - \left(\gamma_i t(k_i) + \gamma_T T(k_i) + \gamma_F T_F(k_i) + \gamma_{-F} T_{-F}(k_i) \right)$$

Routing behaviour

Routing depends on the γ parameters:

- **Egoistic:** $\gamma_i = 1$
- **Social:** $\gamma_T = 1$
- **Selfish fleet:** $\gamma_F = 1$
- **Antisocial / competitive:** $\gamma_T, \gamma_{-F} < 0$



2. Arbitrary utility

Wardropian Utility

$$u_{k,i} \propto t_{k,i}$$

Not anymore

Utility is under the control of the fleet operator and may include:

- $t(k_i)$ — own travel time,
- $T(k_i)$ — system-wide total travel time,
- $T_F(k_i)$ — total fleet travel time,
- $T_{-F}(k_i)$ — total travel time outside the fleet.

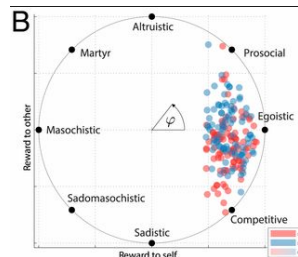
General CAV path utility

$$u_{k,i} = - \left(\gamma_i t(k_i) + \gamma_T T(k_i) + \gamma_F T_F(k_i) + \gamma_{-F} T_{-F}(k_i) \right)$$

Routing behaviour

Routing depends on the γ parameters:

- **Egoistic:** $\gamma_i = 1$
- **Social:** $\gamma_T = 1$
- **Selfish fleet:** $\gamma_F = 1$
- **Antisocial / competitive:** $\gamma_T, \gamma_{-F} < 0$



3. Predictive capabilities → games

Prediction

Accurate predictions of joint actions

$$\mathcal{K} = \{k_i : i \in \mathcal{Q}\}$$

may be a key component of efficient CAV fleet routing.

Single vehicle

A vehicle predicts choices of other users $\mathcal{K}_{-i}$ and resulting network travel times $t_a : a \in A$.

This does not differ much from current real-time **route guidance**.

CAV fleet

A fleet may predict human choices:

$$\mathcal{K}_{-F} = \{k_i : i \notin F\}.$$

Stackelberg Game

The route-choice problem may then become a Stackelberg game: the fleet operator acts as a leader anticipating the actions of human drivers.

Human Adaptation

But humans will react, right?

Competing fleets

Even more interestingly, for the case of competing fleets $F, F', F'', \dots \in \mathcal{F}$ which both predict human actions, but **cannot** predict strategic actions of their competitors.

Leading to intriguing competition patterns, where human drivers are likely to be just a background for a game-theoretical interplay.



3. Predictive capabilities → games

Prediction

Accurate predictions of joint actions

$$\mathcal{K} = \{k_i : i \in Q\}$$

may be a key component of efficient CAV fleet routing.

Single vehicle

A vehicle predicts choices of other users $\mathcal{K}_{-i}$ and resulting network travel times $t_a : a \in A$.

This does not differ much from current real-time **route guidance**.

CAV fleet

A fleet may predict human choices:

$$\mathcal{K}_{-F} = \{k_i : i \notin F\}.$$

Stackelberg Game

The route-choice problem may then become a Stackelberg game: the fleet operator acts as a leader anticipating the actions of human drivers.

Human Adaptation

But humans will react, right?

Competing fleets

Even more interestingly, for the case of competing fleets $F, F', F'', \dots \in \mathcal{F}$ which both predict human actions, but **cannot** predict strategic actions of their competitors.

Leading to intriguing competition patterns, where human drivers are likely to be just a background for a game-theoretical interplay.



3. Predictive capabilities → games

Prediction

Accurate predictions of joint actions

$$\mathcal{K} = \{k_i : i \in \mathcal{Q}\}$$

may be a key component of efficient CAV fleet routing.

Single vehicle

A vehicle predicts choices of other users $\mathcal{K}_{-i}$ and resulting network travel times $t_a : a \in A$.

This does not differ much from current real-time **route guidance**.

CAV fleet

A fleet may predict human choices:

$$\mathcal{K}_{-F} = \{k_i : i \notin F\}.$$

Stackelberg Game

The route-choice problem may then become a Stackelberg game: the fleet operator acts as a leader anticipating the actions of human drivers.

Human Adaptation

But humans will react, right?

Competing fleets

Even more interestingly, for the case of competing fleets $F, F', F'', \dots \in \mathcal{F}$ which both predict human actions, but **cannot** predict strategic actions of their competitors.

Leading to intriguing competition patterns, where human drivers are likely to be just a background for a game-theoretical interplay.



3. Predictive capabilities → games

Prediction

Accurate predictions of joint actions

$$\mathcal{K} = \{k_i : i \in \mathcal{Q}\}$$

may be a key component of efficient CAV fleet routing.

Single vehicle

A vehicle predicts choices of other users $\mathcal{K}_{-i}$ and resulting network travel times $t_a : a \in A$.

This does not differ much from current real-time **route guidance**.

CAV fleet

A fleet may predict human choices:

$$\mathcal{K}_{-F} = \{k_i : i \notin F\}.$$

Stackelberg Game

The route-choice problem may then become a Stackelberg game: the fleet operator acts as a leader anticipating the actions of human drivers.

Human Adaptation

But humans will react, right?

Competing fleets

Even more interestingly, for the case of competing fleets $F, F', F'', \dots \in \mathcal{F}$ which both predict human actions, but **cannot** predict strategic actions of their competitors.

Leading to intriguing competition patterns, where human drivers are likely to be just a background for a game-theoretical interplay.



3. Predictive capabilities → games

Prediction

Accurate predictions of joint actions

$$\mathcal{K} = \{k_i : i \in \mathcal{Q}\}$$

may be a key component of efficient CAV fleet routing.

Single vehicle

A vehicle predicts choices of other users $\mathcal{K}_{-i}$ and resulting network travel times $t_a : a \in A$.

This does not differ much from current real-time **route guidance**.

CAV fleet

A fleet may predict human choices:

$$\mathcal{K}_{-F} = \{k_i : i \notin F\}.$$

Stackelberg Game

The route-choice problem may then become a Stackelberg game: the fleet operator acts as a leader anticipating the actions of human drivers.

Human Adaptation

But humans will react, right?

Competing fleets

Even more interestingly, for the case of competing fleets $F, F', F'', \dots \in \mathcal{F}$ which both predict human actions, but **cannot** predict strategic actions of their competitors.

Leading to intriguing competition patterns, where human drivers are likely to be just a background for a game-theoretical interplay.



4. Route choice or fleet assignment?

Route choice

Given a choice set K_i , expected utilities and predicted travel times, human drivers behaviourally select:

$$k_i = \arg \max_{k \in K_i} u_{k,i}.$$

Fleet assignment

CAV algorithms are not constrained behaviourally. From the fleet perspective, this is no longer necessarily a choice, but an assignment.

Joint action

A fleet controls multiple vehicles, each may choose any path. The action space grows exponentially with fleet size: 50 vehicles and 10 paths yield 10^{50} possible joint actions.

$$|K|^{N|}$$

Optimal fleet assignment

$$\mathcal{K}_F^* = \arg \min_{\mathcal{K}_F \in \prod_{i \in F} K_i} u_F(\mathcal{K}_F, \gamma, \mathcal{K}_{-F}).$$

this assumes we:

- 1 well **predicted** others' choices $-F$ and
- 2 found optimum within own **huge** action space.



4. Route choice or fleet assignment?

Route choice

Given a choice set K_i , expected utilities and predicted travel times, human drivers behaviourally select:

$$k_i = \arg \max_{k \in K_i} u_{k,i}.$$

Fleet assignment

CAV algorithms are not constrained behaviourally. From the fleet perspective, this is no longer necessarily a choice, but an assignment.

Joint action

A fleet controls multiple vehicles, each may choose any path. The action space grows exponentially with fleet size: 50 vehicles and 10 paths yield 10^{50} possible joint actions.

$$|K|^{|N|}$$

Optimal fleet assignment

$$\mathcal{K}_F^* = \arg \min_{\mathcal{K}_F \in \prod_{i \in F} K_i} u_F(\mathcal{K}_F, \gamma, \mathcal{K}_{-F}).$$

this assumes we:

- 1 well **predicted** others' choices $-F$ and
- 2 found optimum within own **huge** action space.



4. Route choice or fleet assignment?

Route choice

Given a choice set K_i , expected utilities and predicted travel times, human drivers behaviourally select:

$$k_i = \arg \max_{k \in K_i} u_{k,i}.$$

Fleet assignment

CAV algorithms are not constrained behaviourally. From the fleet perspective, this is no longer necessarily a choice, but an assignment.

Joint action

A fleet controls multiple vehicles, each may choose any path. The action space grows exponentially with fleet size: 50 vehicles and 10 paths yield 10^{50} possible joint actions.

$$|K|^{10}$$

Optimal fleet assignment

$$\mathcal{K}_F^* = \arg \min_{\mathcal{K}_F \in \prod_{i \in F} K_i} u_F(\mathcal{K}_F, \gamma, \mathcal{K}_{-F}).$$

this assumes we:

- 1 well **predicted** others' choices $-F$ and
- 2 found optimum within own **huge** action space.



Fleet strategy

beyond utility

Possible objectives - Uber case

The strategy may optimise:

- routing efficiency,
- profit,
- market share.

Fleet strategy

A joint action κ_F^* realizes fleet strategy S_F if:

$$\kappa_F^* = \arg \max_{\kappa_F \in \prod_{i \in F} K_i} \sum_{\tau \in T} S_F(\kappa_F).$$

possibly over multiple days of evolution (long-term effect)

For instance: market maximizer

The probability $\pi_{F,i}$ of joining fleet F may depend on:

- 1 individual perception (exogenous),
- 2 costs $c_{F,i}$ (control) and
- 3 the travel-time advantage ($\sim \gamma, \kappa_F$):

$$\Delta t_{i,F} = \bar{t}_{i,-F} - \bar{t}_{i,F}.$$

how much faster than competitor are you? fleet controls both ($\gamma_{-F} < 0$)

Strategy

$$\pi_{F,i} \propto (\beta_{F,i}, c_{F,i}, \Delta t_{i,F}, \dots).$$

The fleet strategy may therefore result from an interplay between routing efficiency, individual characteristics and pricing.



Fleet strategy

beyond utility

Possible objectives - Uber case

The strategy may optimise:

- routing efficiency,
- profit,
- market share.

Fleet strategy

A joint action κ_F^* realizes fleet strategy S_F if:

$$\kappa_F^* = \arg \max_{\kappa_F \in \prod_{i \in F} K_i} \sum_{\tau \in T} S_F(\kappa_F).$$

possibly over multiple days of evolution (long-term effect)

For instance: market maximizer

The probability $\pi_{F,i}$ of joining fleet F may depend on:

- 1 individual perception (exogenous),
- 2 costs $c_{F,i}$ (control) and
- 3 the travel-time advantage ($\sim \gamma, \kappa_F$):

$$\Delta t_{i,F} = \bar{t}_{i,-F} - \bar{t}_{i,F}.$$

how much faster than competitor are you? fleet controls both ($\gamma_{-F} < 0$)

Strategy

$$\pi_{F,i} \propto (\beta_{F,i}, c_{F,i}, \Delta t_{i,F}, \dots).$$

The fleet strategy may therefore result from an interplay between routing efficiency, individual characteristics and pricing.



Fleet strategy

beyond utility

Possible objectives - Uber case

The strategy may optimise:

- routing efficiency,
- profit,
- market share.

Fleet strategy

A joint action κ_F^* realizes fleet strategy S_F if:

$$\kappa_F^* = \arg \max_{\kappa_F \in \prod_{i \in F} K_i} \sum_{\tau \in T} S_F(\kappa_F).$$

possibly over multiple days of evolution (long-term effect)

For instance: market maximizer

The probability $\pi_{F,i}$ of joining fleet F may depend on:

- 1 individual perception (exogenous),
- 2 costs $c_{F,i}$ (control) and
- 3 the travel-time advantage ($\sim \gamma, \kappa_F$):

$$\Delta t_{i,F} = \bar{t}_{i,-F} - \bar{t}_{i,F}.$$

how much faster then competitor are you? fleet controls both ($\gamma_{-F} < 0$)

Strategy

$$\pi_{F,i} \propto (\beta_{F,i}, c_{F,i}, \Delta t_{i,F}, \dots).$$

The fleet strategy may therefore result from an interplay between routing efficiency, individual characteristics and pricing.

Post-Wardropian framework



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

```

1: for each day  $d$  do
2:   for each fleet  $F \in \mathcal{F}$  do
3:      $k_i : i \in -F \leftarrow k_i$ 
4:      $k_i : i \in \mathcal{F} \setminus Fs$ 
5:      $S * F$ 
6:      $C_{F,i}$ 
7:      $\mathcal{K}_F$ 
8:   end for
9:   for each agent  $i$  do
10:     $\mathcal{K}_d \leftarrow k_i$ 
11:   end for
12:    $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$ 
13:    $C_d(\mathcal{K})$ 
14:   for each fleet  $F \in \mathcal{F}$  do
15:     $S_F$ 
16:   end for
17:   for each agent  $i$  do
18:     $\bar{t}_{k,i}$ 
19:     $F_i + = i$ 
20:   end for
21: end for
22: return

```

- ▷ day-to-day simulation
- ▷ fleets act independently
- ▷ predict human choices
 - ▷ predict other fleets
 - ▷ pick own strategy
- ▷ set prices for individual users
- ▷ assign fleet vehicles to routes
- ▷ decision (humans) or assignment (CAVs)
- ▷ simulate the system
- ▷ measure the system performance
 - ▷ fleet learning
- ▷ Learn outcomes of applying the fleet strategy
 - ▷ human learning/daptation
 - ▷ update cost expectation (learn/adapt)
 - ▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus FS$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus FS$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus Fs$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S * F$

6: $C_{F,i}$

7: $\mathcal{K}_F$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: S_F

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k,i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

```

1: for each day  $d$  do
2:   for each fleet  $F \in \mathcal{F}$  do
3:      $k_i : i \in -F \leftarrow k_i$ 
4:      $k_i : i \in \mathcal{F} \setminus FS$ 
5:      $S * F$ 
6:      $C_{F,i}$ 
7:      $\mathcal{K}_F$ 
8:   end for
9:   for each agent  $i$  do
10:     $\mathcal{K}_d \leftarrow k_i$ 
11:  end for
12:   $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$ 
13:   $C_d(\mathcal{K})$ 
14:  for each fleet  $F \in \mathcal{F}$  do
15:     $S_F$ 
16:  end for
17:  for each agent  $i$  do
18:     $\bar{t}_{k,i}$ 
19:     $F_i + = i$ 
20:  end for
21: end for
22: return
```

- ▷ day-to-day simulation
- ▷ fleets act independently
- ▷ predict human choices
 - ▷ predict other fleets
 - ▷ pick own strategy
- ▷ set prices for individual users
- ▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

- ▷ simulate the system
- ▷ measure the system performance
- ▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

- ▷ human learning/daptation
- ▷ update cost expectation (learn/adapt)
 - ▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S *_{\mathcal{F}}$

6: $C_{\mathcal{F}, i}$

7: $\mathcal{K}_{\mathcal{F}}$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: $S_{\mathcal{F}}$

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k, i}$

19: $F_i += i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S *_{\mathcal{F}}$

6: $C_{\mathcal{F}, i}$

7: $\mathcal{K}_{\mathcal{F}}$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: $S_{\mathcal{F}}$

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k, i}$

19: $F_i += i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

1: **for** each day d **do**

2: **for** each fleet $F \in \mathcal{F}$ **do**

3: $k_i : i \in -F \leftarrow k_i$

4: $k_i : i \in \mathcal{F} \setminus F_S$

5: $S *_{\mathcal{F}}$

6: $C_{\mathcal{F}, i}$

7: $\mathcal{K}_{\mathcal{F}}$

8: **end for**

9: **for** each agent i **do**

10: $\mathcal{K}_d \leftarrow k_i$

11: **end for**

12: $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$

13: $C_d(\mathcal{K})$

14: **for** each fleet $F \in \mathcal{F}$ **do**

15: $S_{\mathcal{F}}$

16: **end for**

17: **for** each agent i **do**

18: $\bar{t}_{k, i}$

19: $F_i + = i$

20: **end for**

21: **end for**

22: **return**

▷ day-to-day simulation

▷ fleets act independently

▷ predict human choices

▷ predict other fleets

▷ pick own strategy

▷ set prices for individual users

▷ assign fleet vehicles to routes

▷ decision (humans) or assignment (CAVs)

▷ simulate the system

▷ measure the system performance

▷ fleet learning

▷ Learn outcomes of applying the fleet strategy

▷ human learning/daptation

▷ update cost expectation (learn/adapt)

▷ joining/leaving Fleet



Algorithm 1: Post-wardropian traffic assignment framework

Require: network $G(N, A)$, demand Q , traffic flow model $\mathcal{M}$

```

1: for each day  $d$  do
2:   for each fleet  $F \in \mathcal{F}$  do
3:      $k_i : i \in -F \leftarrow k_i$ 
4:      $k_i : i \in \mathcal{F} \setminus FS$ 
5:      $S *_{\mathcal{F}}$ 
6:      $C_{\mathcal{F}, i}$ 
7:      $\mathcal{K}_{\mathcal{F}}$ 
8:   end for
9:   for each agent  $i$  do
10:     $\mathcal{K}_d \leftarrow k_i$ 
11:   end for
12:    $t(k_i : i \in Q) \leftarrow \mathcal{M}(\mathcal{K}_d)$ 
13:    $C_d(\mathcal{K})$ 
14:   for each fleet  $F \in \mathcal{F}$  do
15:     $S_{\mathcal{F}}$ 
16:   end for
17:   for each agent  $i$  do
18:     $\bar{t}_{k, i}$ 
19:     $F_i + = i$ 
20:   end for
21: end for
22: return

```

- ▷ day-to-day simulation
- ▷ fleets act independently
- ▷ predict human choices
 - ▷ predict other fleets
 - ▷ pick own strategy
- ▷ set prices for individual users
- ▷ assign fleet vehicles to routes
- ▷ decision (humans) or assignment (CAVs)
- ▷ simulate the system
- ▷ measure the system performance
 - ▷ fleet learning
- ▷ Learn outcomes of applying the fleet strategy
 - ▷ human learning/daptation
 - ▷ update cost expectation (learn/adapt)
 - ▷ joining/leaving Fleet



Limitations

Modelling such cases requires several components:

- a method for selecting optimal strategy to reach the fleet operator's objective,
- an algorithm identifying a fleet **assignment** maximising that strategy,
- a model anticipating competitors' moves,
- a calibrated and accurate route-choice model,
- a human adaptation model.

Gap

Most remain open research problems with no established methods and algorithms.



Limitations

Modelling such cases requires several components:

- a method for selecting optimal strategy to reach the fleet operator's objective,
- an algorithm identifying a fleet **assignment** maximising that strategy,
- a model anticipating competitors' moves,
- a calibrated and accurate route-choice model,
- a human adaptation model.

Gap

Most remain open research problems with no established methods and algorithms.



Equilibria in a Post-Wardropian system

What could a new emerging **equilibria** or stable-states be?

Human drivers

Routing may be considered stable when drivers do not change their day-to-day routing probabilities or choices.

Fleet membership

Fleet may be considered stable as long as member do not reconsider leaving the fleet or joining another fleet.

Fleet operators

A fleet remains in equilibrium when it does not adjust its strategy. **It does not mean it has stable routing day-to-day!**

Traffic system

The system is stable when distributions of traffic flows and travel times do not vary day-to-day.

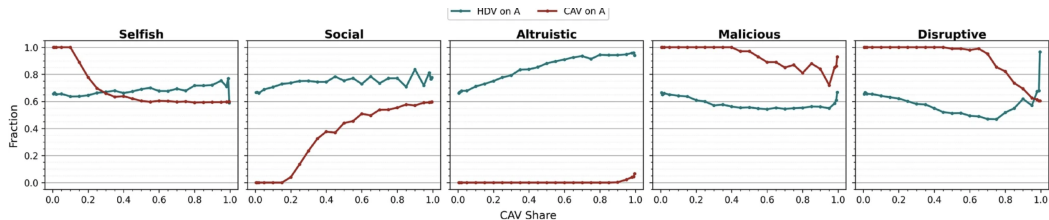
Results



Different choices

Studying the simplest of the two-route bottleneck macroscopic network, we discover¹ that:

CAV route choices (blue-ish) differ significantly from the choices of the remaining human driven vehicles (brown-ish).



optimal CAV choices depend on the market share (x-axis) and the strategy (panels).

¹Jamróz, G., Akman, A. O., Psarou, A., Varga, Z. G. & Kucharski, R. Social implications of coexistence of CAVs and human drivers in the context of route choice. [Scientific Reports](#) **15**, 6768 (2025).

How to include state of the art Machine Learning in the loop?

tool

```
pip install routerl
```

RouteRL: MARL framework for urban route choice with autonomous vehicles

AVs are soon to make routing decisions in our cities.

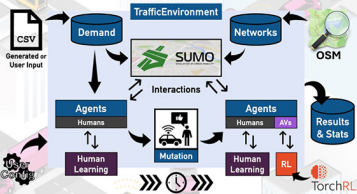
When you become a passenger in your car, what happens to:








To understand better, we present **RouteRL**.



RouteRL provides:

-  A tool to derive practical insights into future traffic dynamics.
-  A highly customizable experimental ground for scientific validation.
-  A challenging testbed for evaluating RL solutions.

Ahmet Onur Akman*, Anastasia Psarou*, Łukasz Gorczyca, Zoltán György Varga, Grzegorz Jamróz, Rafał Kucharski



This research is financed by the European Union within the Horizon Europe Framework Programme, ERC Starting Grant number 101075838: COeXISTENCE.

2

²Akman, A. O., Psarou, A., Gorczyca, Ł., Varga, Z. G., Jamróz, G. & Kucharski, R. RouteRL: Multi-agent reinforcement learning framework for urban route choice with autonomous vehicles. [SoftwareX 31, 102279](https://doi.org/10.1016/j.softx.2025.102279). ISSN: 2352-7110. doi:<https://doi.org/10.1016/j.softx.2025.102279> (2025).

URB Urban Routing Benchmark

Leaderboard

URB benchmarks routing policies for mixed traffic with human drivers and connected autonomous vehicles (CAVs). The table below indexes experiment folders under results/ and ranks runs by benchmark metrics.

434 runs indexed

[Join the leaderboard](#)

[Home](#)
[Leaderboard](#)
[Studies](#)

Filters

ENV CONFIG

☒ clusters_at1
 ☒ clusters_at2
 ☒ clusters_sumo_obs
 ☒ config_ing_short
 ☒ config_pnx_short
 ☒ config1
 ☒ config1_eta2
 ☒ config1_smoothing_off

ENV SEED

☒ 40
 ☒ 41
 ☒ 42
 ☒ 43
 ☒ 44

TASK CONFIG

☒ altruistic_40
 ☒ altruistic_50
 ☒ collaborative_40
 ☒ collaborative_50
 ☒ collaborative_60
 ☒ collectivist_40
 ☒ collectivist_50
 ☒ collectivist_60
 ☒ competitive_40

[ingolstadt_custom2](#)
[proville](#)
[ingolstadt_custom](#)
[saint_arnault](#)

Experiment ID	Contributors	Script	Env config	Task config	Algorithm Config	Env seed	Torch Seed	t_pre	t_test	t_train	t_CAV	t_HDV_pre	t_HDV_test	CAV_advantage	Eff
#1 tbf_ing_short_s...	👤👤👤	iqj_tabular.py	config_ing_short	selfish_40	config_ing_short	varies	varies	4.501675	4.401656	4.523914	4.280405	4.48794	4.48249	1.047463	1.0
#2 tbf_ing2_c3				competitive_40	config1_10k	44	varies	4.514129	4.413811	4.413195	4.334315	4.463233	4.467394	1.030913	1.0
#3 tbf_ing2_c3				social_40	config1_10k	44	varies	4.514129	4.413989	4.428907	4.334258	4.463233	4.467343	1.030659	1.0
#4 tbf_ing2_c3				selfish_40	config1_10k	44	varies	4.514129	4.414857	4.418046	4.336513	4.463233	4.467087	1.03011	1.0
#5 tbf_ing2_c3				collaborative_40	config1_10k	44	varies	4.514129	4.41603	4.429173	4.337871	4.463233	4.468136	1.03003	1.0
#6 tbf_ing2_c3				collaborative_40	config1_longer	44	varies	4.514129	4.41064	4.467645	4.340726	4.463233	4.45725	1.026845	1.0
#7 iqj_ing_clust				competitive_40	config1_longer	44	varies	4.872278	4.789435	4.828549	4.713607	4.821161	4.839987	1.026813	1.0

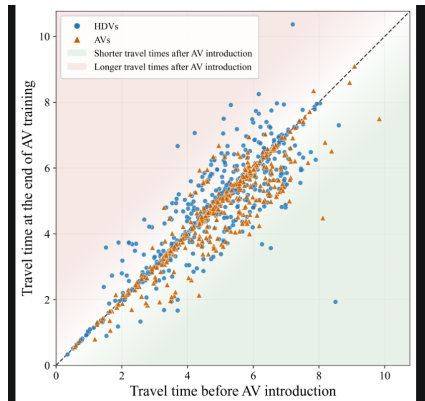
³Akman, A. O., Psarou, A., Hoffmann, M., Gorczyca, Ł., Kowalski, Ł., Gora, P., Jamróz, G. & Kucharski, R. URB Urban Routing Benchmark for RL-equipped Connected Autonomous Vehicles. [NeurIPS 2025](#). eprint: <https://doi.org/10.26434/chemrxiv-2025-10000>

PostWardropian, R. Kucharski, hEART 2026 Paris

49

Ingolstadt

40% CAVs, SUMO, Selfish, IQL, 3000 episodes training



Travel time change after introducing AV routing for HDVs (blues) and CAVs (orange).

Conclusions



Conclusions

- complex phenomena ahead
- largely unknown
- challenging to model
- macroscopic? rather not.
- analytical? rather ML, black-box deep learning
- equilibria? more of a chaotic evolution.
- a lot to do for us, **or LLMs...**

Thank you



Rafał Kucharski

Jagiellonian University, Kraków

rafal.kucharski@uj.edu.pl

Acknowledgements

This research was funded by the COeXISTENCE project (grant number 101075838), which is financed by the European Research Council.

